Data Analysis in Contour

Lab Guide

Contents

1. Use case

About the Company

TitaniumWorks Manufacturing is a company that produces high-quality industrial equipment. Like most manufacturing organizations, it relies heavily on machinery that must operate efficiently and safely. To maintain smooth operations, the company regularly inspects its equipment to prevent breakdowns and costly delays.

The Challenge

The maintenance team is struggling to decide which machines should be inspected next. The process is currently manual and time-consuming, leading to inefficiencies and missed priorities. As a data analyst, you have been asked to step in and use Contour to bring data-driven insights into the decision-making process.
With Contour, you can:

- Save time by automatically re-running analyses when new data arrives.
- Build complex analyses easily through a visual, point-and-click interface.
- Create clear visualizations and reports that help non-technical teams understand results and act quickly.

Your Goal

In this exercise, your main objective is to support TitaniumWorks in improving its equipment inspection strategy through data analysis. You will:

1. Clean and prepare the datasets for accurate analysis.
2. Combine multiple datasets to create a complete view of equipment performance.
3. Add calculated fields to reveal insights about maintenance schedules and identify high-risk machines.
4. Design visual dashboards to make the findings easy to interpret.
5. Embed your visualizations in a Notepad for seamless collaboration and presentation.

Datasets

In this Lab, you'll be working with two datasets - one that stores equipment data and one that stores the production data of parts that are produced by the equipment.

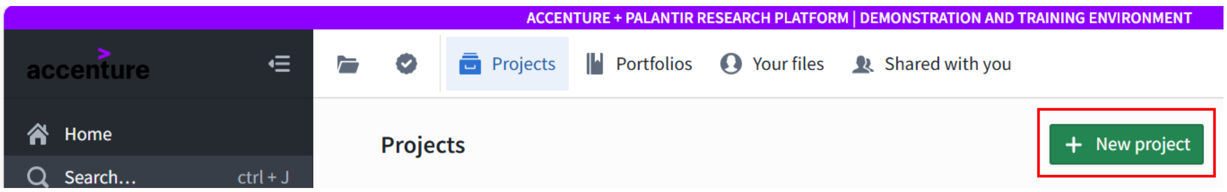
a. equipment.csv
b. parts.csv

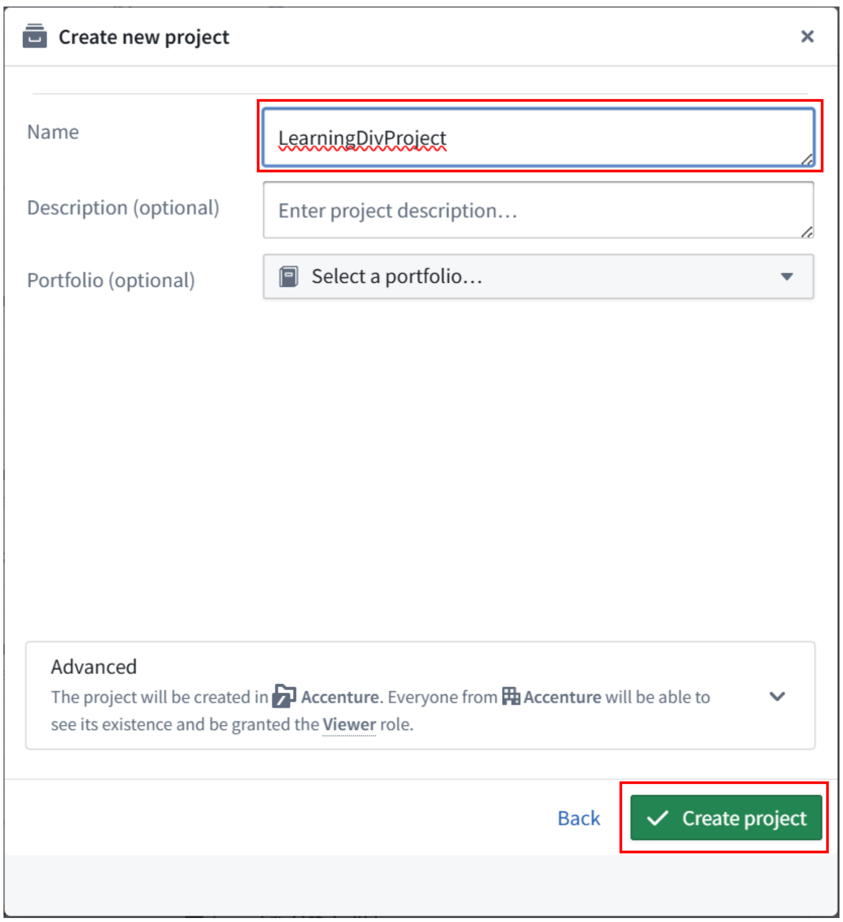
2. Project Setup for Contour

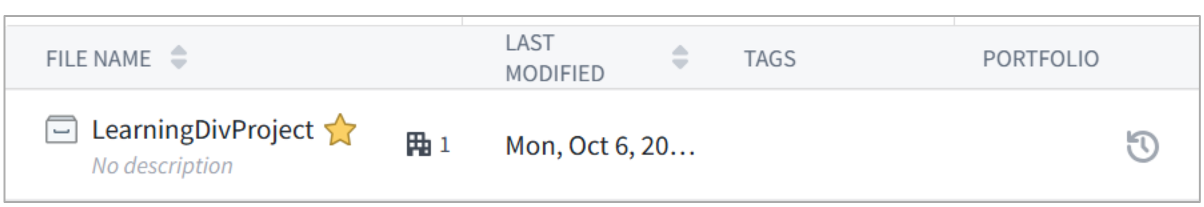
- If you already have your own Learning Project on your Foundry instance you can skip this step and move on to the next step to create your course-specific folder.

a. Create a Project

Navigate to Home Page -> Click on New Project -> Choose Basic Project under Template -> Provide a Name -> Click on Create Project

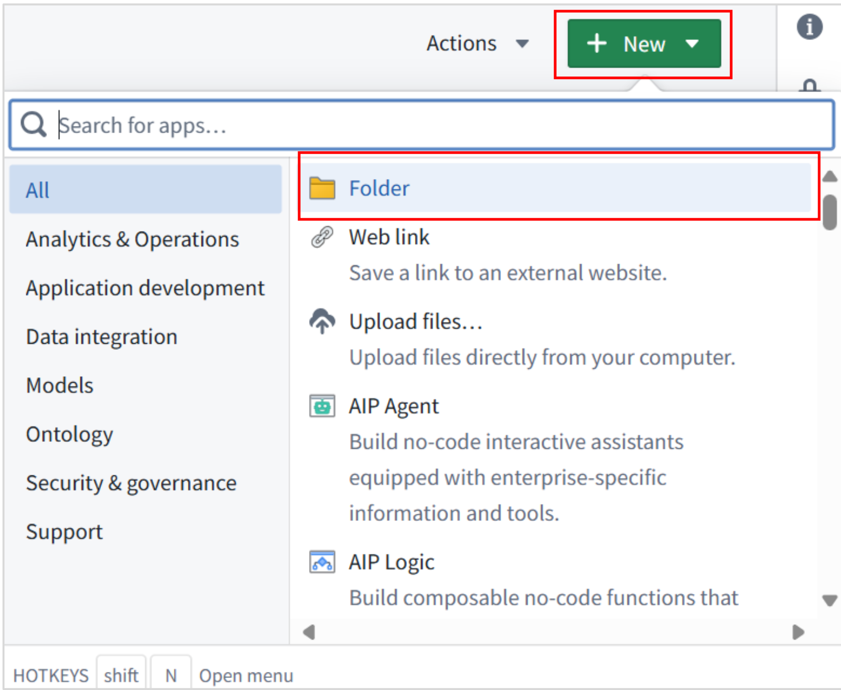


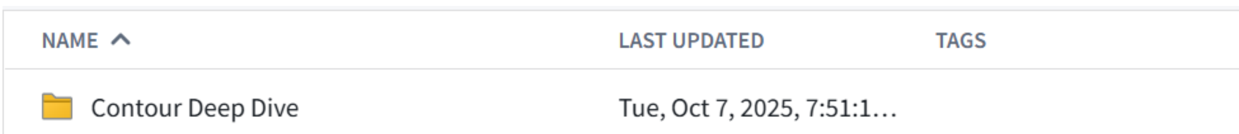




b. Create a Folder

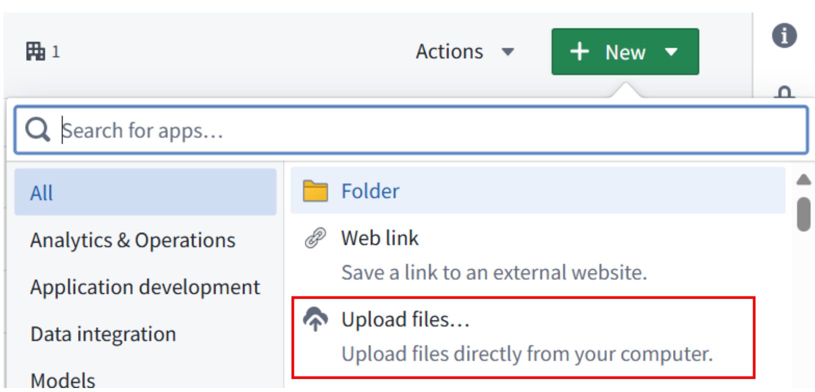
Open the Project folder -> Click on New -> Choose Folder -> Name the folder as Contour Deep Dive



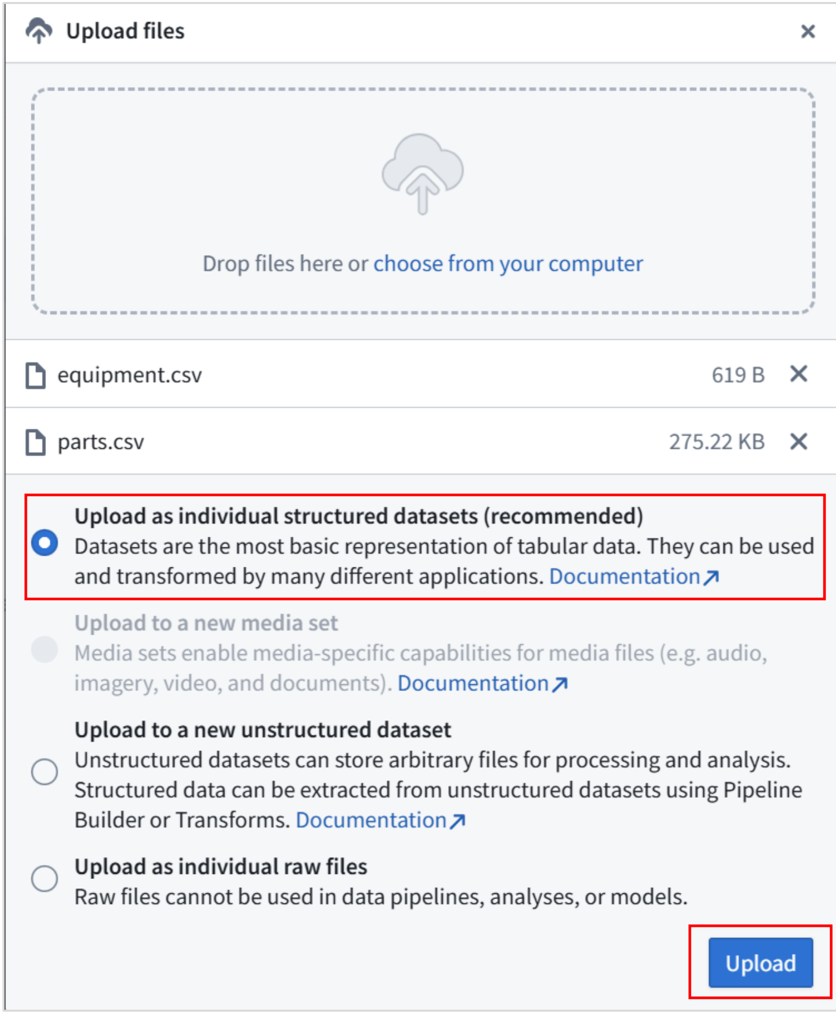


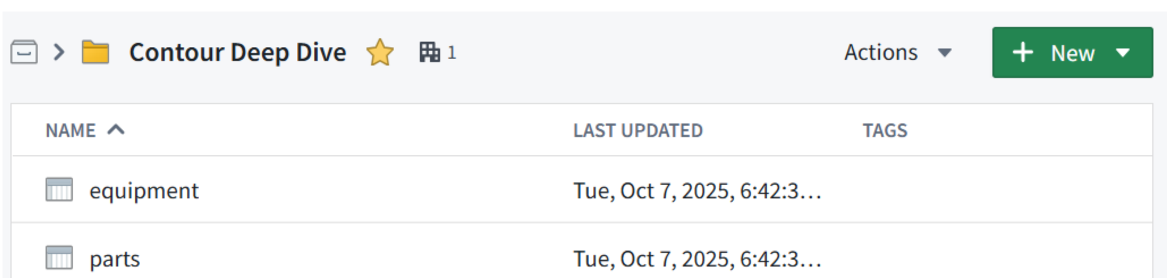
3. Upload Data

a. Open the Contour Deep Dive Folder -> Click New -> Choose upload files



b. Click on Choose from your computer -> Select the files -> Click on open -> Choose Upload as individual structured datasets -> click on Upload

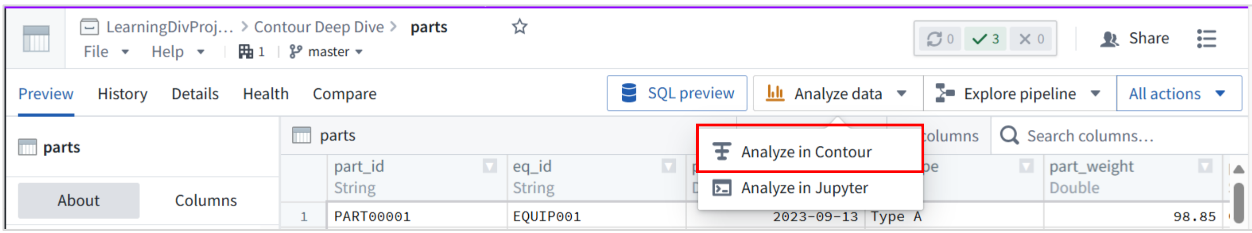


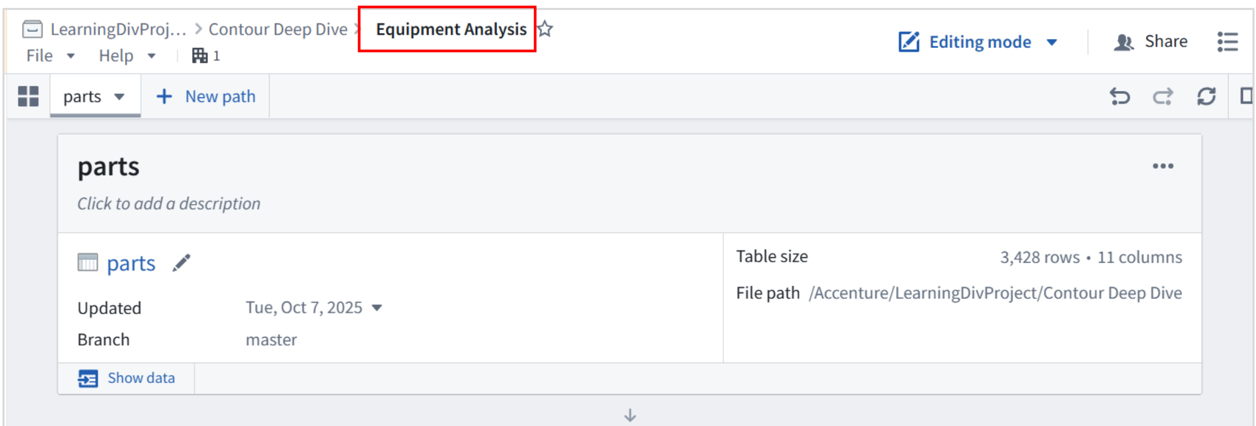


Upon successful uploading you can view the datasets under the Contour Deep Dive folder.

4. Create a Contour Analysis

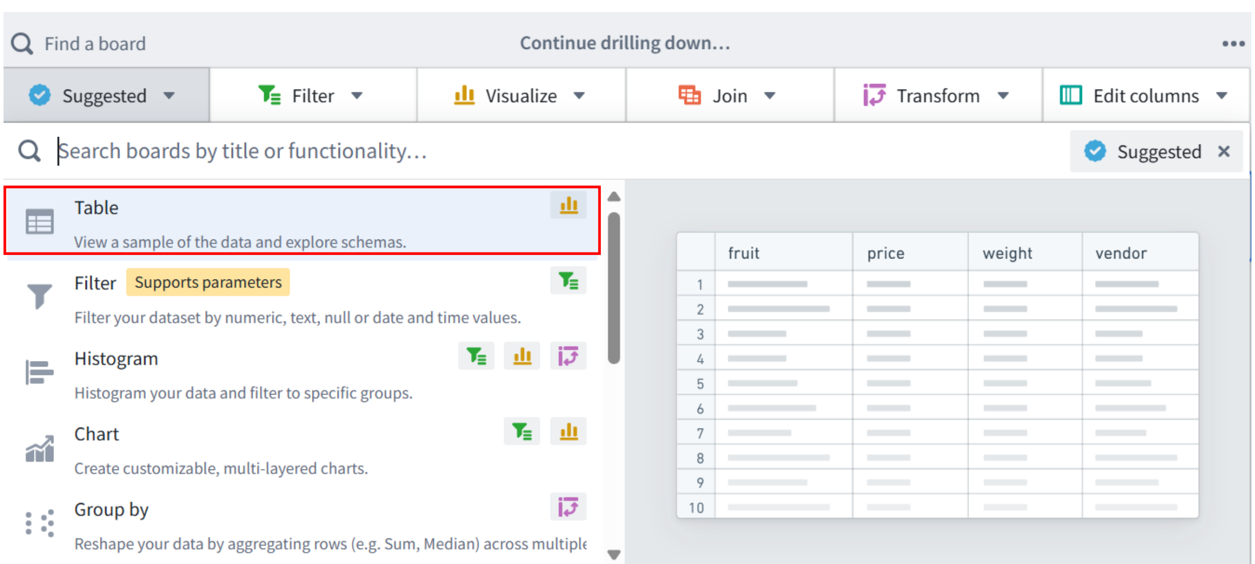
➢ Select Parts file -> Click in Analyze Data -> Choose Analyze in Contour -> Rename the analysis as Equipment Analysis



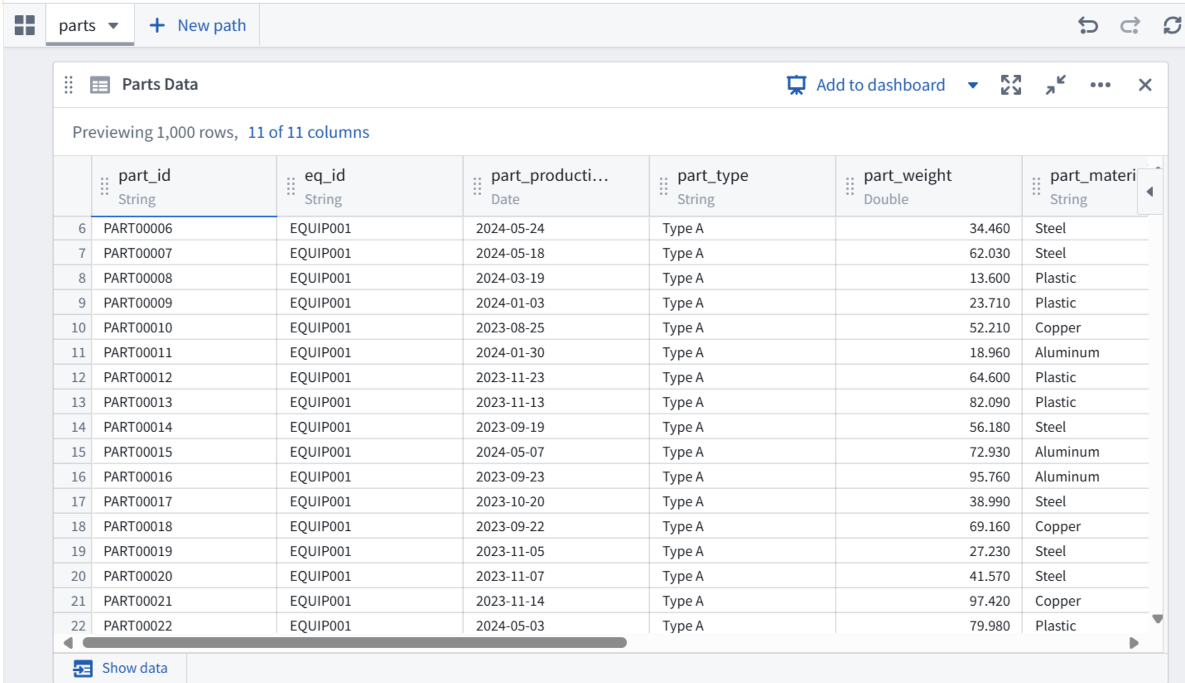


5. Table Board

➢ In the board toolbar, click Suggested -> Table



Click on show data to display the data



	part_id String	eq_id String	part_producti… Date	part_type String	part_weight Double	part_materi String
6	PART00006	EQUIP001	2024-05-24	Type A	34.460	Steel
7	PART00007	EQUIP001	2024-05-18	Type A	62.030	Steel
8	PART00008	EQUIP001	2024-03-19	Type A	13.600	Plastic
9	PART00009	EQUIP001	2024-01-03	Type A	23.710	Plastic
10	PART00010	EQUIP001	2023-08-25	Type A	52.210	Copper
11	PART00011	EQUIP001	2024-01-30	Type A	18.960	Aluminum
12	PART00012	EQUIP001	2023-11-23	Type A	64.600	Plastic
13	PART00013	EQUIP001	2023-11-13	Type A	82.090	Plastic
14	PART00014	EQUIP001	2023-09-19	Type A	56.180	Steel
15	PART00015	EQUIP001	2024-05-07	Type A	72.930	Aluminum
16	PART00016	EQUIP001	2023-09-23	Type A	95.760	Aluminum
17	PART00017	EQUIP001	2023-10-20	Type A	38.990	Steel
18	PART00018	EQUIP001	2023-09-22	Type A	69.160	Copper
19	PART00019	EQUIP001	2023-11-05	Type A	27.230	Steel
20	PART00020	EQUIP001	2023-11-07	Type A	41.570	Steel
21	PART00021	EQUIP001	2023-11-14	Type A	97.420	Copper
22	PART00022	EQUIP001	2024-05-03	Type A	79.980	Plastic

6. Calculation Board

In the board toolbar, click Visualize

Calculation 1: Calculate the number of unique parts by choosing the

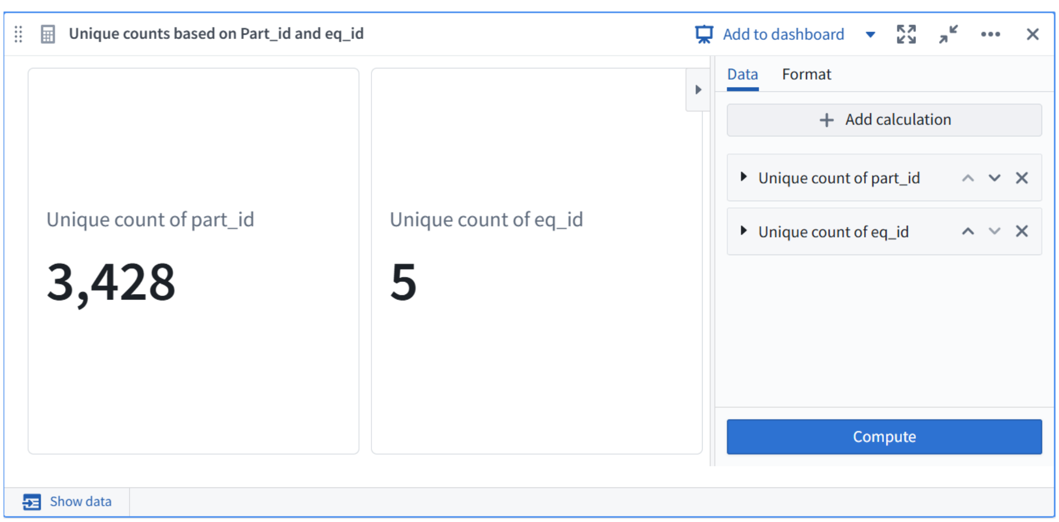
Column: part_id

Aggregate: Unique count

Calculation 2: Calculate the number of unique equipment ID by choosing the

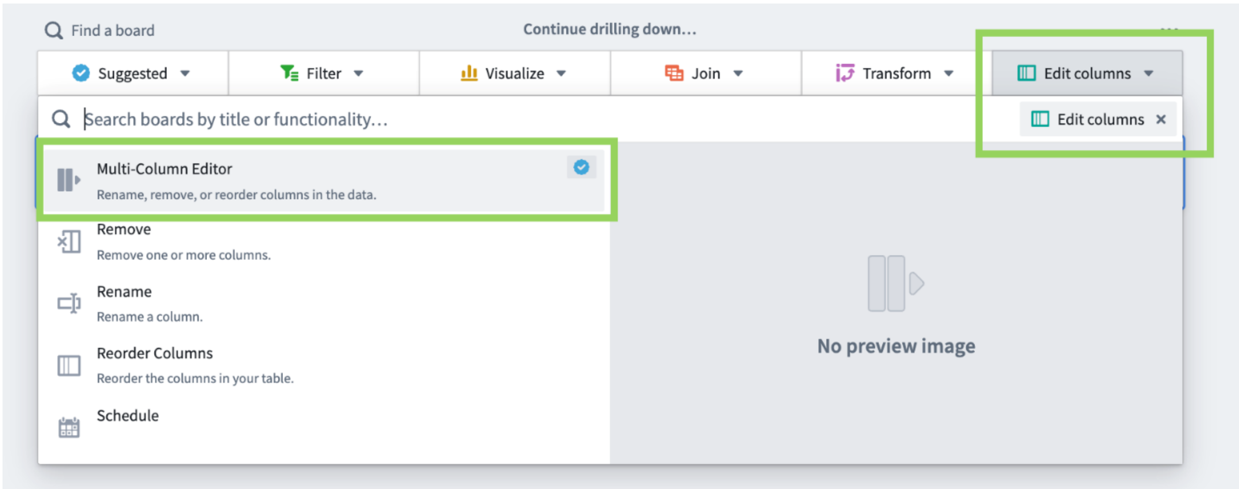
Column: eq_id

Aggregate: Unique count

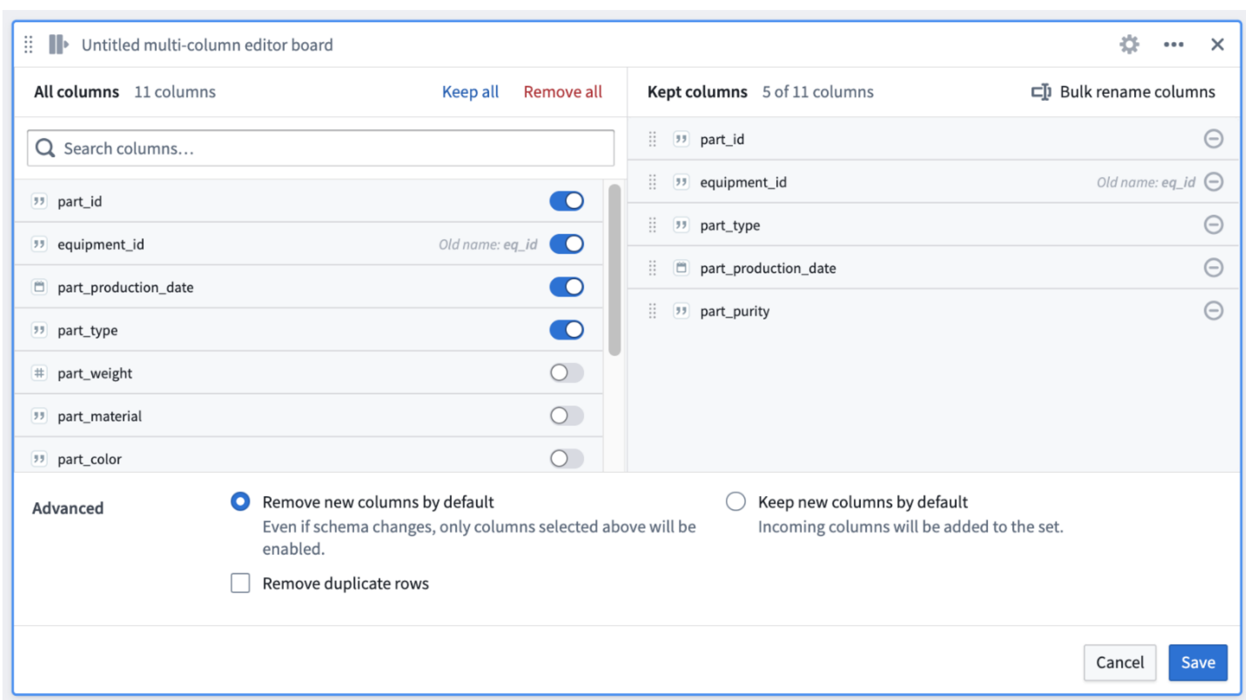


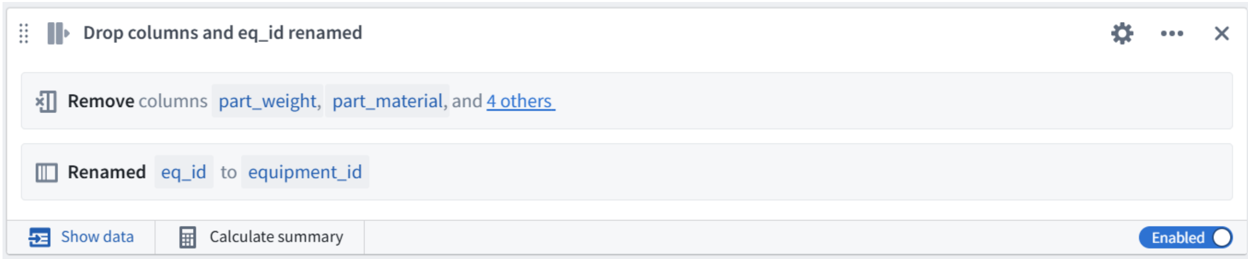
7. Rename and Drop Columns

- In the board toolbar, Edit columns > Multi-Column Editor



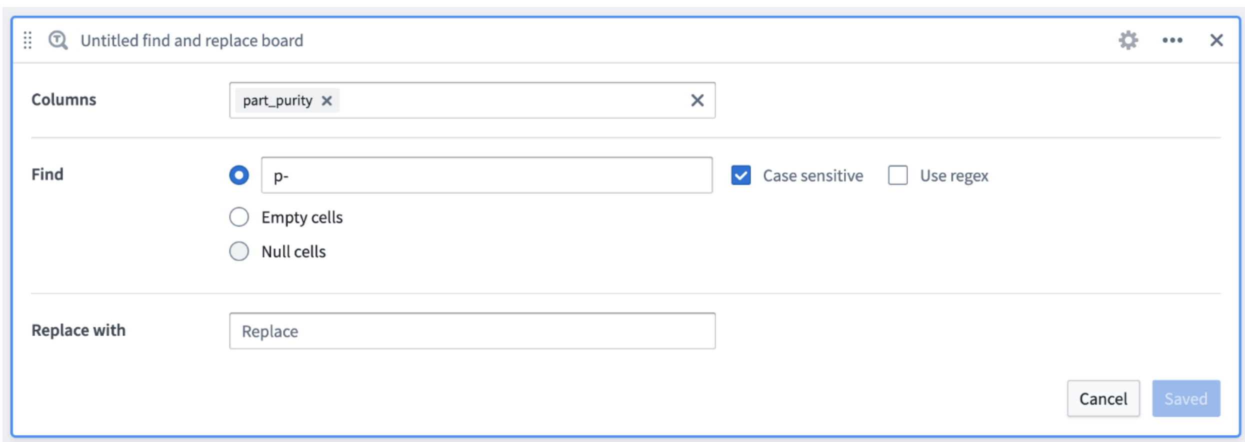
- In the right Kept columns panel, click on the eq_id column -> rename the column to equipment_id
- Under all columns, toggle off the below columns and click apply / save
  - part_weight
  - part_material
  - part_color
  - part_dimensions
  - part_cost
  - part_quality_grade

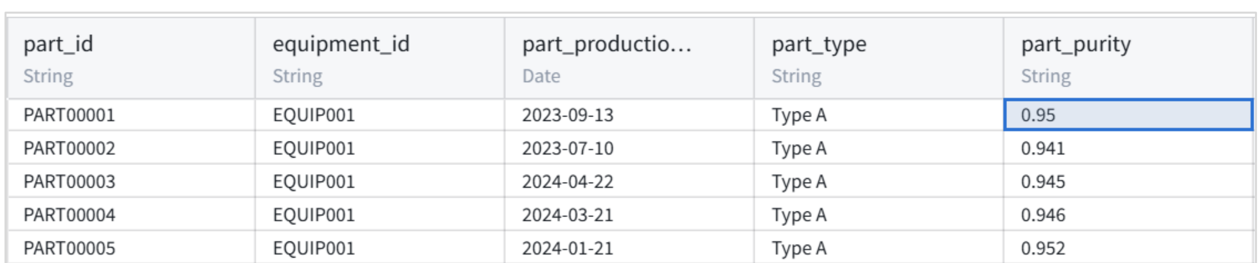




8. Find and Replace Column Board

- In the board toolbar, **Transforms > Find and Replace** -> Remove the p- prefix from part_purity using below options and click on Apply
    - Columns: part_purity
    - Find: p-
    - Replace: (leave this empty)

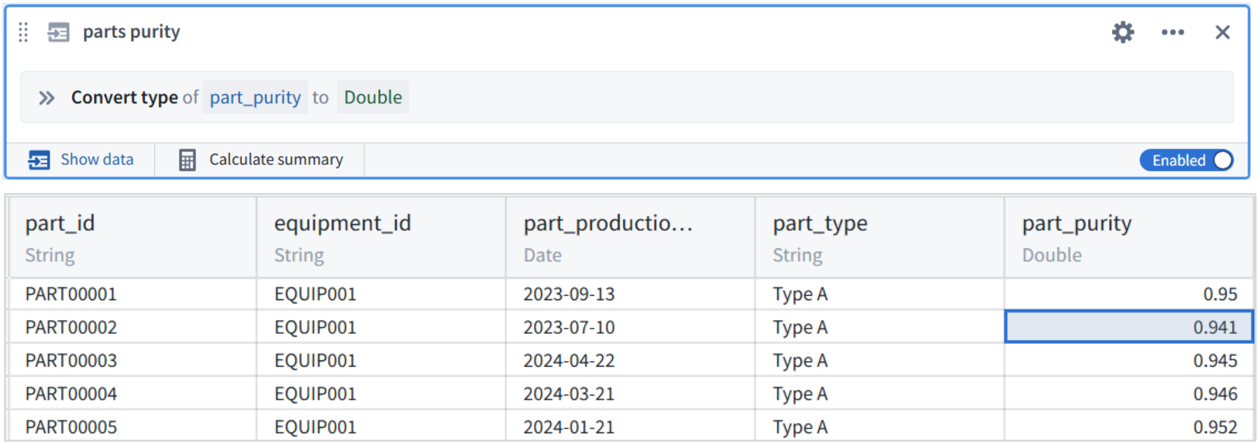




part_id String	equipment_id String	part_productio... Date	part_type String	part_purity String
PART00001	EQUIP001	2023-09-13	Type A	0.95
PART00002	EQUIP001	2023-07-10	Type A	0.941
PART00003	EQUIP001	2024-04-22	Type A	0.945
PART00004	EQUIP001	2024-03-21	Type A	0.946
PART00005	EQUIP001	2024-01-21	Type A	0.952

9. Convert Types Board

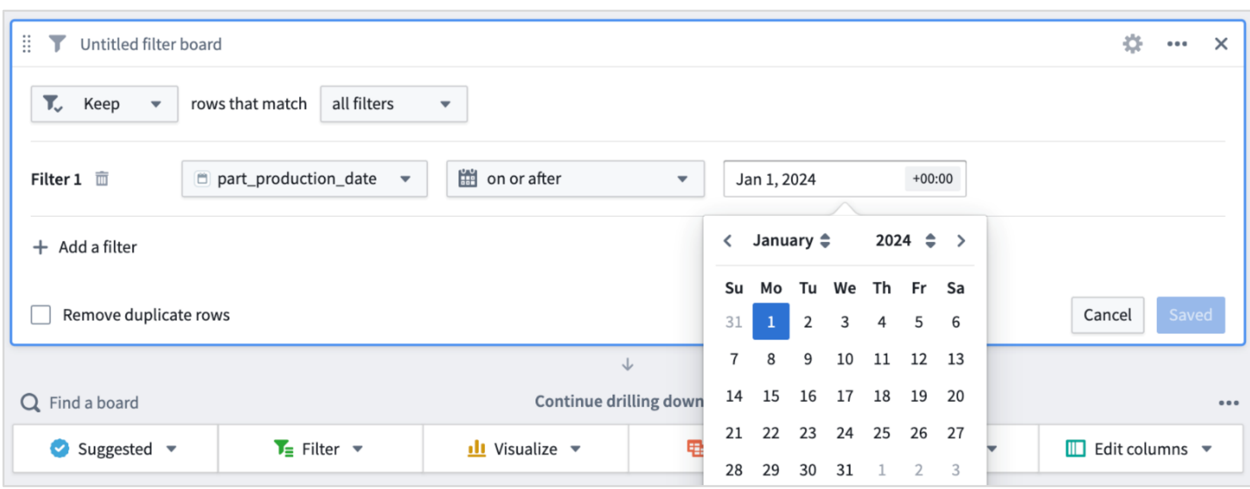
- In the board toolbar, **Transform > Convert types** -> convert the part_purity to Double by using the below options and click on Apply
    - Type conversions: Select column > part_purity
    - On the right, change String to Double

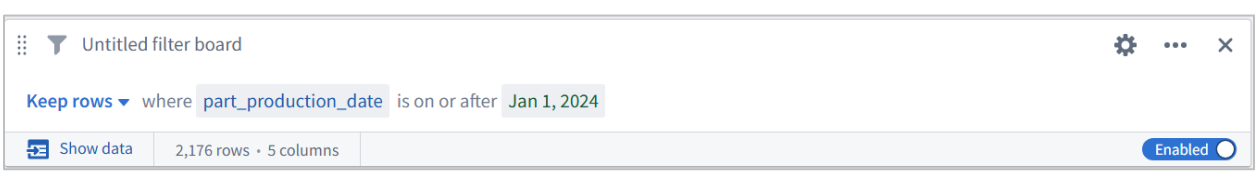


10. Filter Board

➢ In the board toolbar, **Filter > Filter.** Set Filter 1 to keep only production dates in 2024 using the below options and click Apply.

  - Select column: part_production_date
  - Comparison: on or after
  - Date: Select Jan 1, 2024





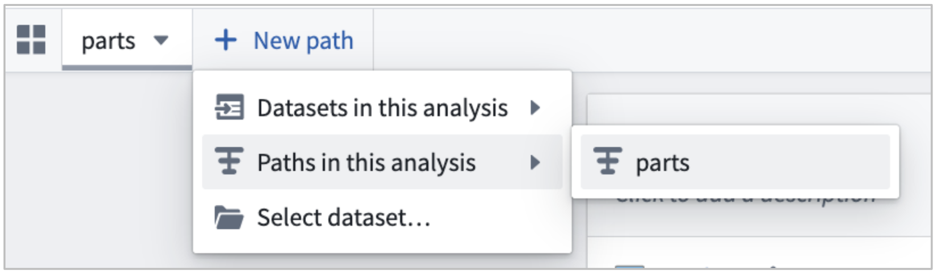
Click **Calculate summary** in the lower left of your new filter board

- After calculation, you should see a row count of 2,176 rows. All subsequent boards will use this filtered data.

11. Joining Datasets

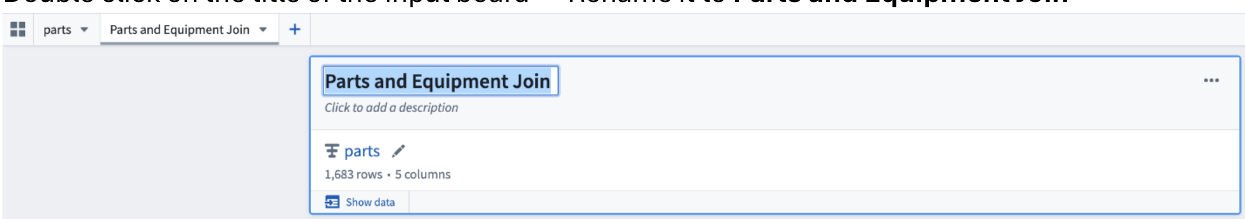
a. Add New Path to the Analysis

➢ Click on the new path at the top of the analysis -> Choose Paths in the analysis -> parts



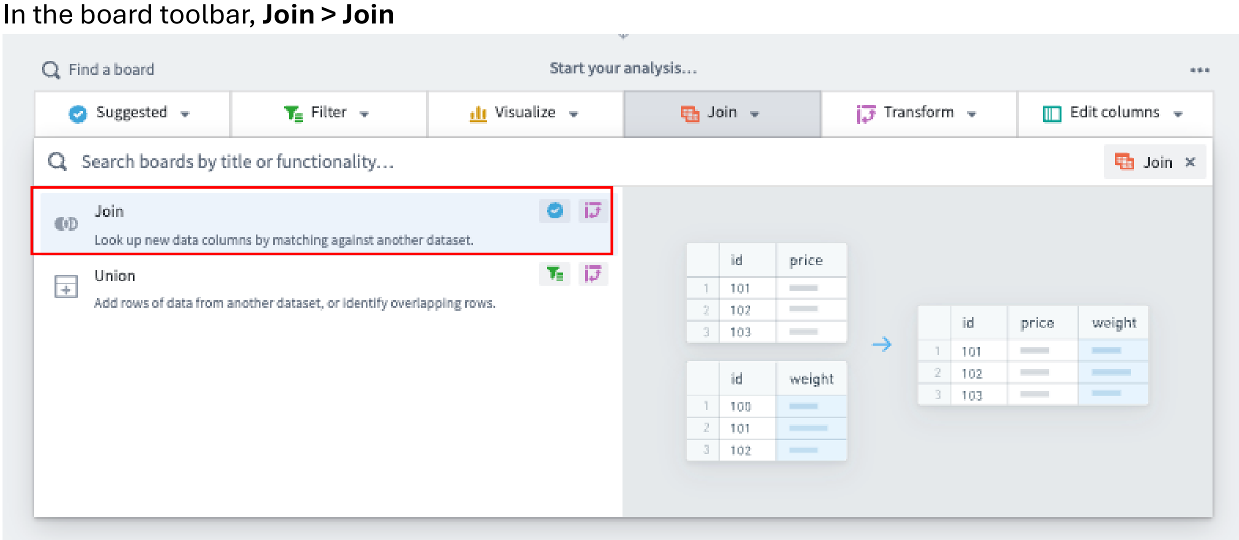
In the new tab, the first input board will display **parts (resulting set)** as the title. This means the output of the **parts** path will flow as input into this new path

➢ Double click on the title of the input board -> Rename it to **Parts and Equipment Join**

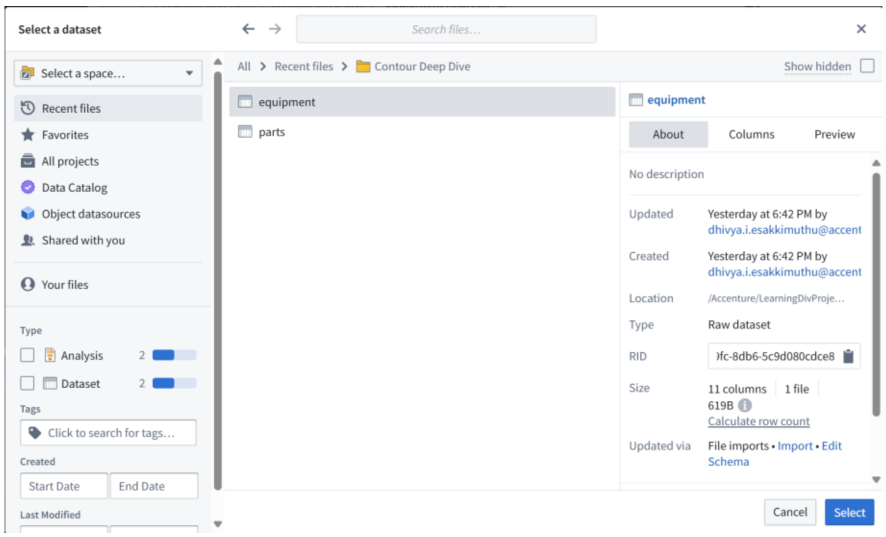


b. Joining Datasets

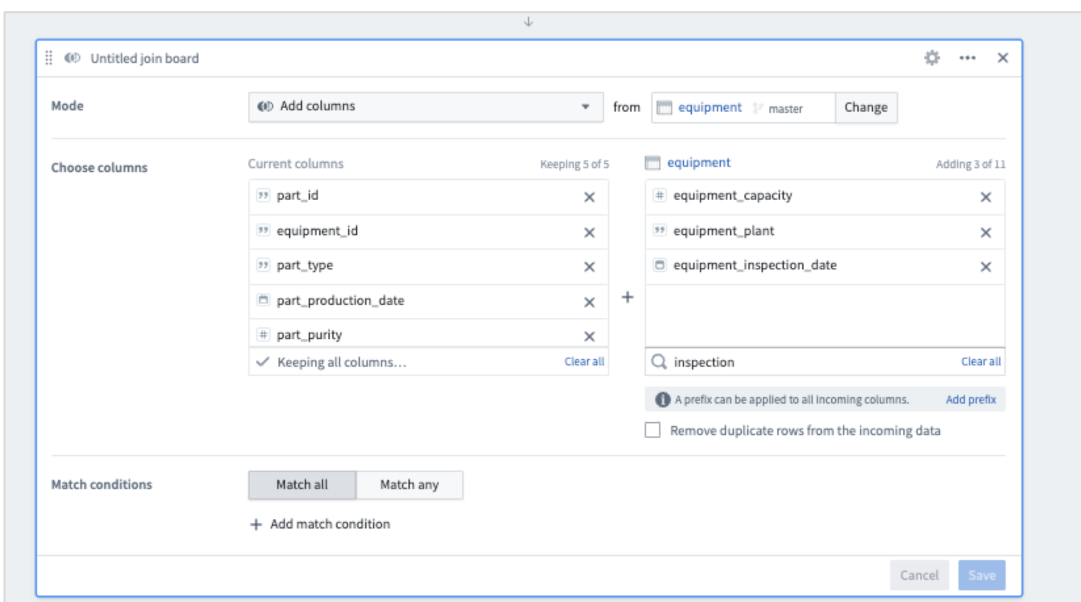
➢ In the board toolbar, **Join > Join**



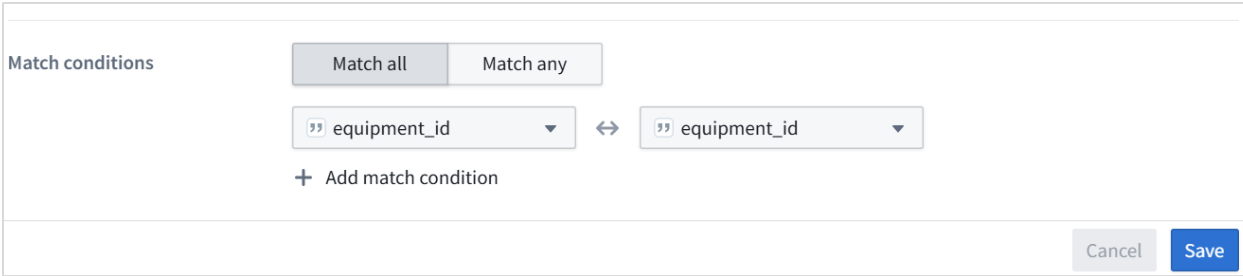
- Add columns from the equipment dataset
  - Leave the Mode set to Add columns
  - Click Choose dataset… -> Select dataset
  - Search for and select the equipment dataset that you previously uploaded to your course folder



  - In the equipment field on the right side of the window, select Add columns to add
    - equipment_capacity, equipment_plant, equipment_inspection_date
  - *Warning: There is both an installation and an inspection date.*



- Add Matching condition as below and click on Save

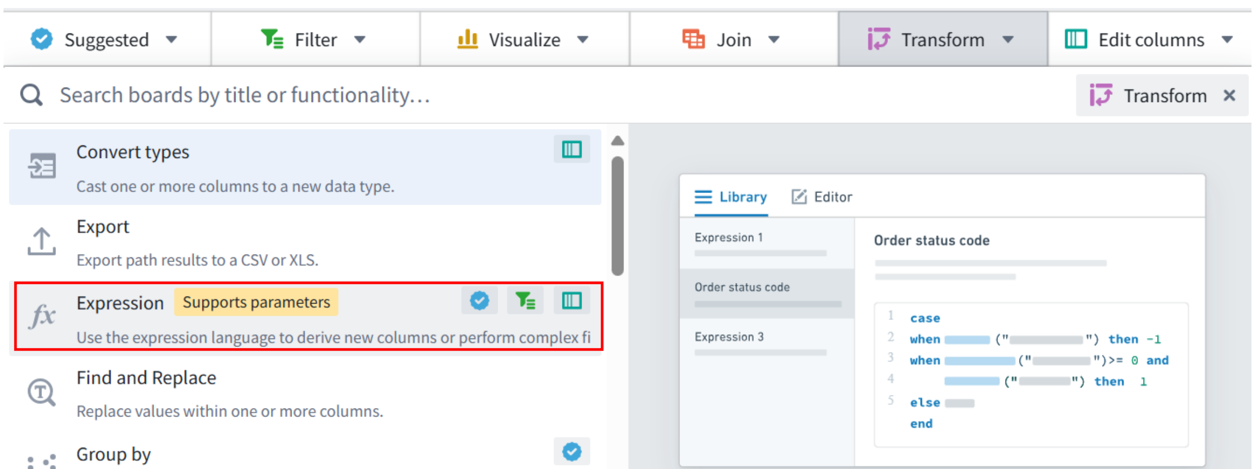


Click on show data to display the data.

equipment_id String	part_productio... Date	part_type String	part_purity Double	equipment_ca... Integer	equipment_plant String
EQUIP001	2024-04-22	Type A	0.945	1200	Plant 1
EQUIP001	2024-03-21	Type A	0.946	1200	Plant 1
EQUIP001	2024-01-21	Type A	0.952	1200	Plant 1
EQUIP001	2024-05-24	Type A	0.954	1200	Plant 1
EQUIP001	2024-05-18	Type A	0.943	1200	Plant 1

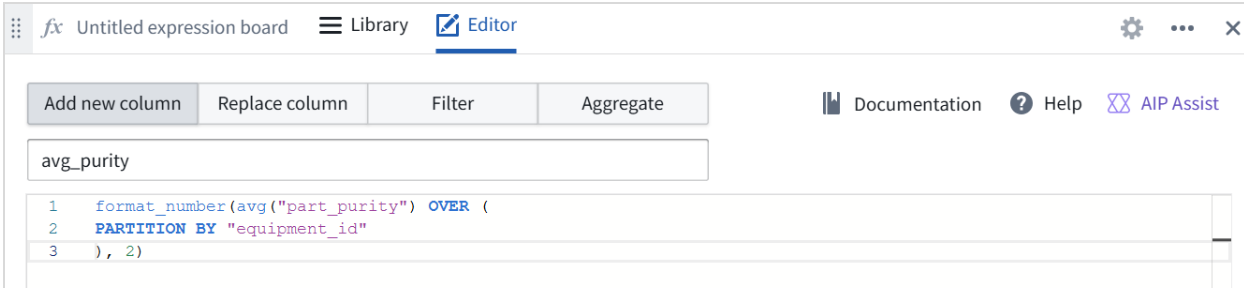
12. Expression Board

- In the board toolbar, Transform > Expression -> Click the Editor tab



- Create a new column using the expression below -> click on save

format_number(avg("part_purity") OVER (
PARTITION BY "equipment_id"
), 2)



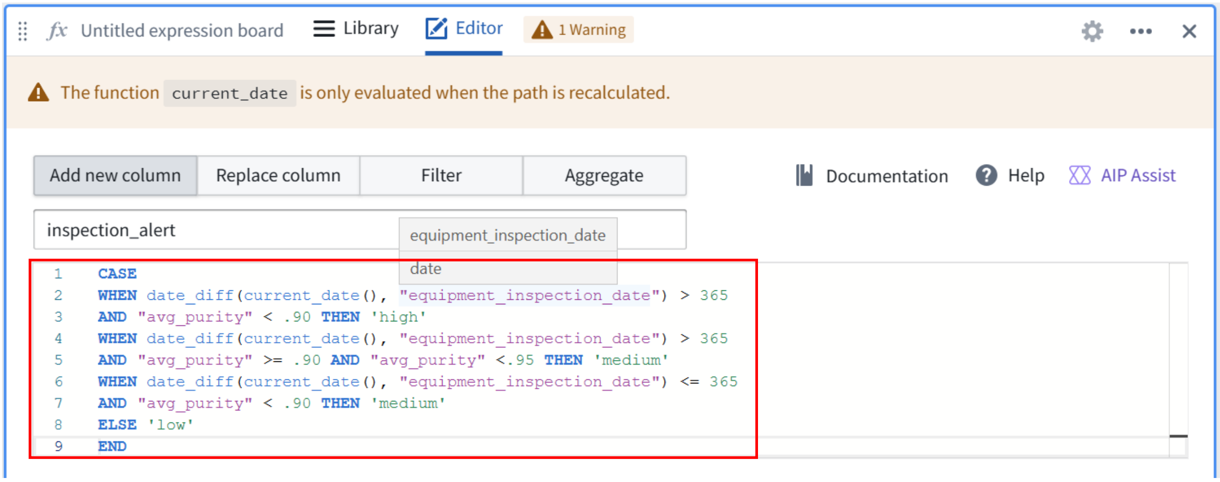
Click on show data to verify the expression value.

avg_purity String	part_id String	equipment_id String	part_productio… Date	part_type String	part_purity Double
0.95	PART00003	EQUIP001	2024-04-22	Type A	0.945
0.95	PART00004	EQUIP001	2024-03-21	Type A	0.946
0.95	PART00005	EQUIP001	2024-01-21	Type A	0.952
0.95	PART00006	EQUIP001	2024-05-24	Type A	0.954
0.95	PART00007	EQUIP001	2024-05-18	Type A	0.943

13. Expression board for Case

- Use another Expression board to generate inspection alerts -> Add another **Expression** board (**Transform > Expression**) -> Under Editor tab -> Add New column named **inspection_alert**.
- Use the below case expression and click on save

```
CASE
WHEN date_diff(current_date(), "equipment_inspection_date") > 365
AND "avg_purity" < .90 THEN 'high'
WHEN date_diff(current_date(), "equipment_inspection_date") > 365
AND "avg_purity" >= .90 AND "avg_purity" <.95 THEN 'medium'
WHEN date_diff(current_date(), "equipment_inspection_date") <= 365
AND "avg_purity" < .90 THEN 'medium'
ELSE 'low'
END
```



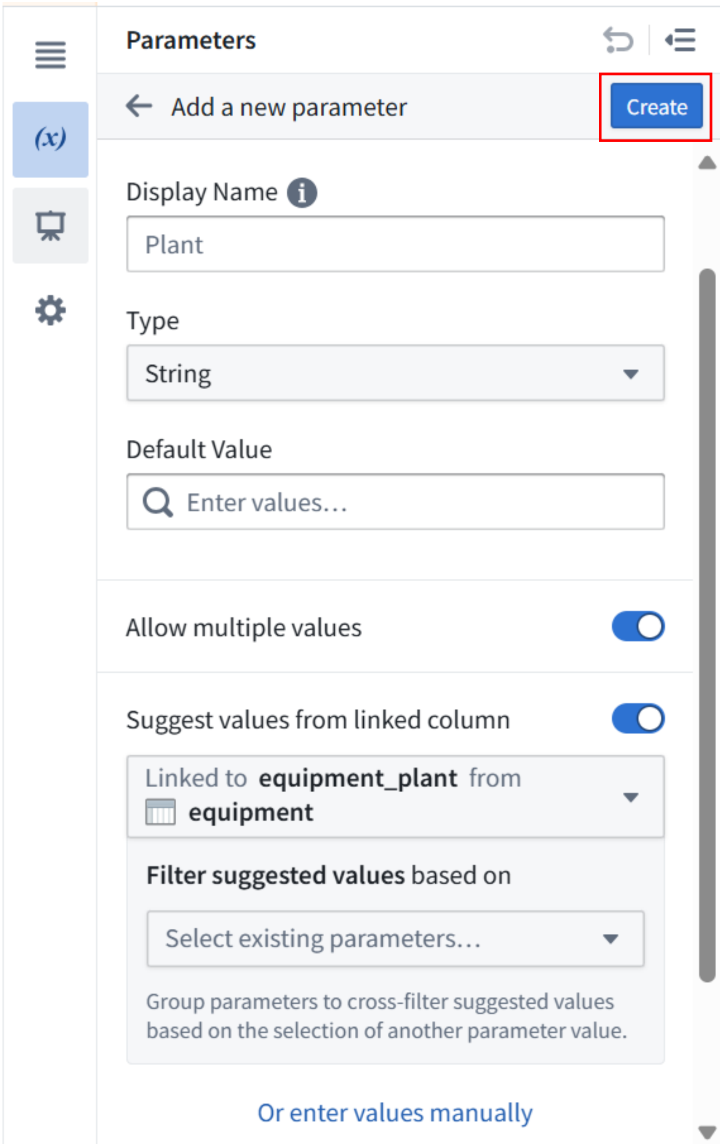
Click on show data to preview the results

inspection_alert String	avg_purity String	part_id String	equipment_id String	part_productio... Date	part_type String
low	0.95	PART00681	EQUIP001	2024-04-19	Type A
low	0.95	PART00682	EQUIP001	2024-03-30	Type A
low	0.95	PART00683	EQUIP001	2024-02-03	Type A
high	0.89	PART00685	EQUIP002	2024-01-06	Type B
high	0.89	PART00688	EQUIP002	2024-03-05	Type B

14. Parameters

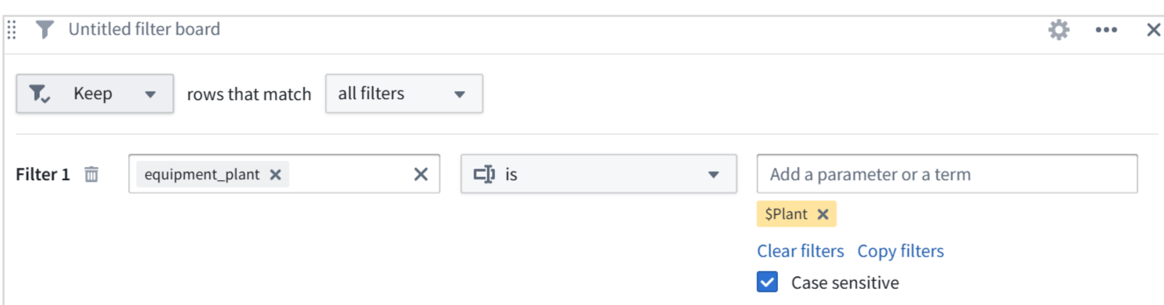
a. Create a parameter

- Click on the **Parameters** icon ***(x)*** from the left-hand sidebar
- Click the + Create parameter button at the center of the left panel.
- Configure the parameter as below and click on create



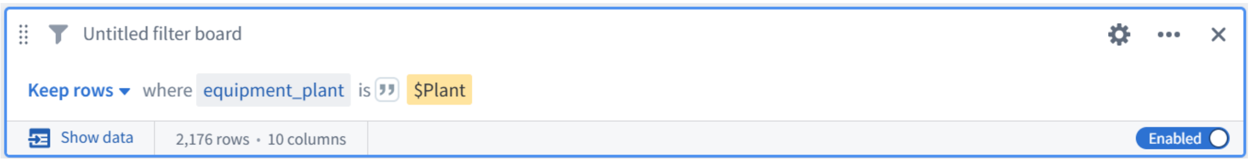
b. Parameter in filter

- From the board toolbar, **Filter > Filter**
- Define as filter below with the parameter and click on save

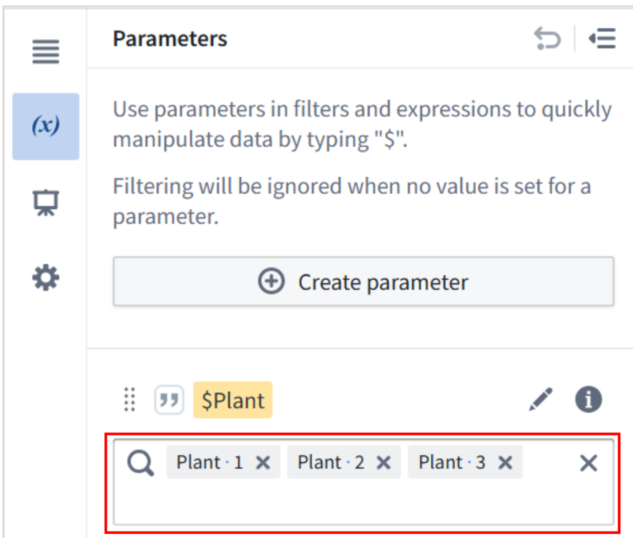


c. Pass values to Parameters

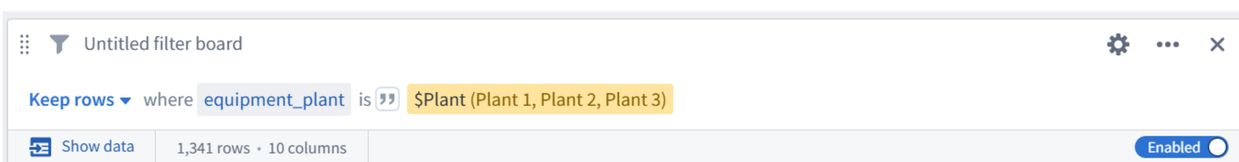
- Click Calculate summary in the lower left corner of the two most recent expression boards
  - Notice that the datasets are the same at this point. You have not defined the value for the $plant parameter, so no filtering is being done.



- Open the left-side **Parameters** panel -> Enter values for **$plant**: Plant 1, Plant 2, and Plant 3
- Click **Apply** at the bottom right of the Parameters panel



- Now click **Calculate summary** on your latest filter board
- The row count should update to 1,341 as you are now filtering only to rows containing data for Plant 1, Plant 2, and Plant 3.

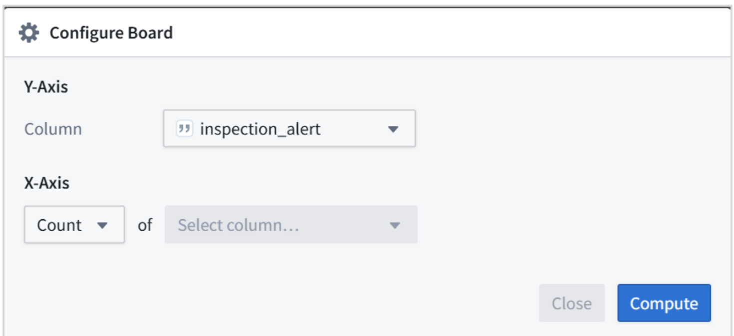


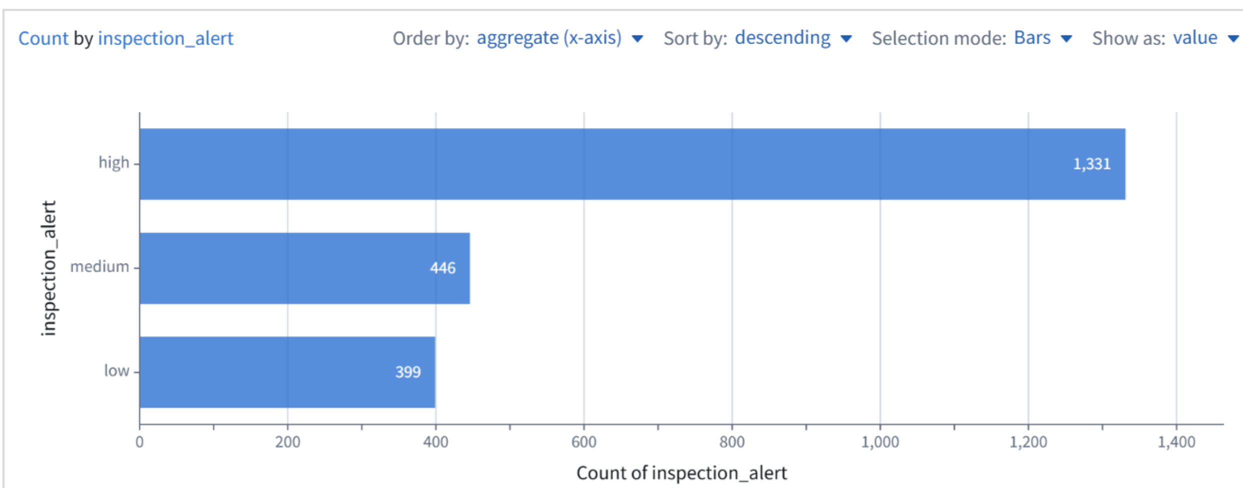
Clear the parameters value in the parameter panel and click on apply before proceeding with the next step

15. Visualizations

a. Histogram

- In the board toolbar, **Visualize > Histogram** -> Configure the board to show a count of inspection alerts using the below options and click on compute
    - Under Y-Axis, Column: inspection_alert
    - Under X-Axis, select Count



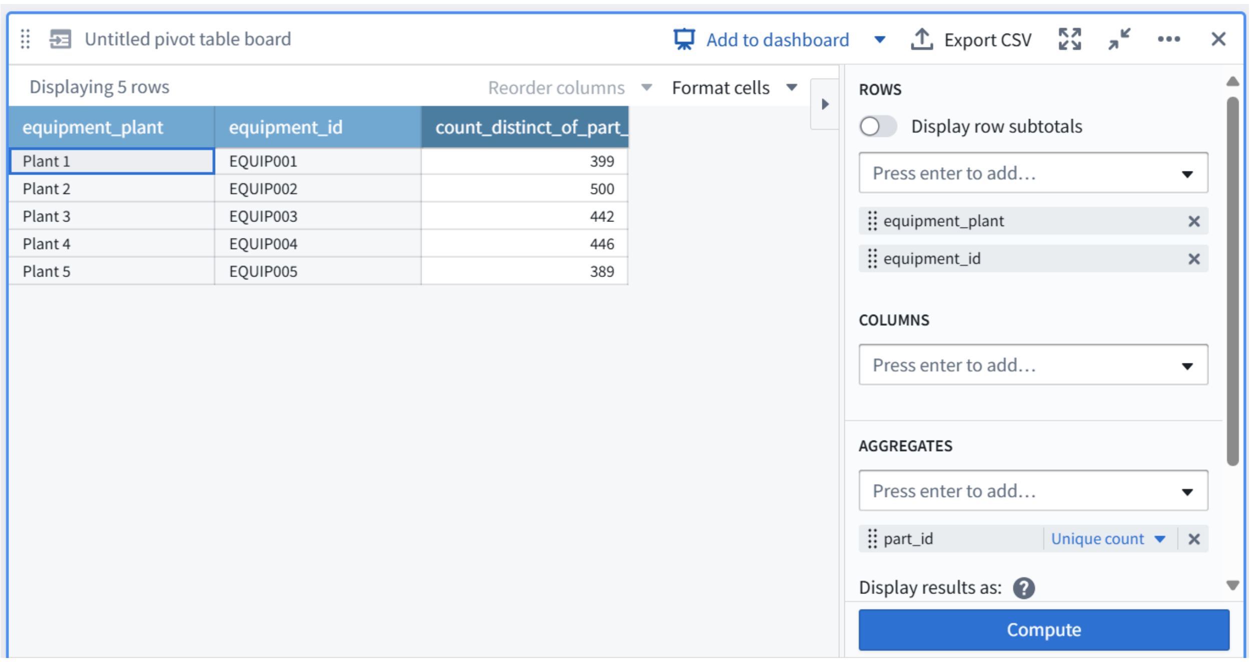


Click on the bar representing high values.

A grey bar at the bottom of the board will appear displaying, "Keep rows where inspection_alert is high." This will filter the data in all subsequent boards.

b. Pivot Table

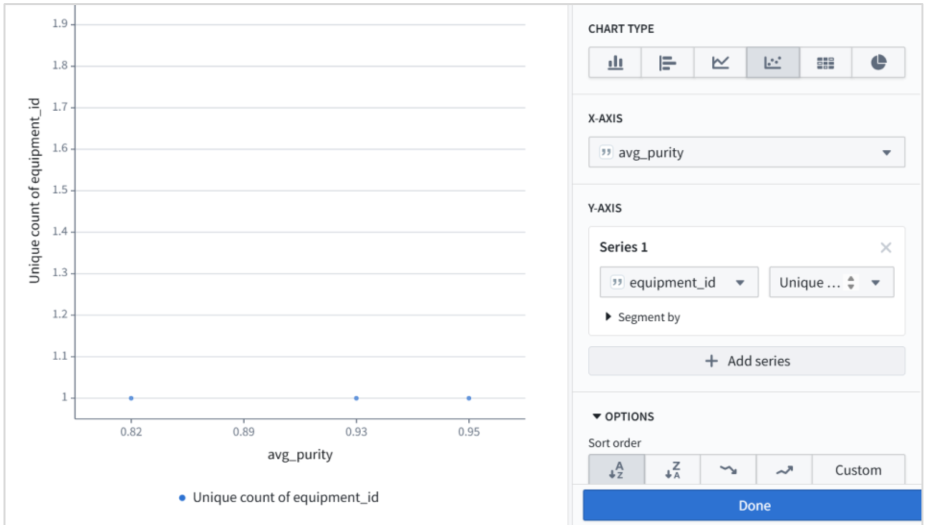
- In the board toolbar, **Transform > Pivot Table** -> Configure the pivot table as below and click on compute
  - Rows: Add equipment_plant and equipment_id
  - Columns: Leave empty
    - While you have the option to select a column for a deeper analysis of each aggregated row, we'll leave it blank for this simple analysis.
  - Aggregates: Select part_id and ensure that Unique count is selected.



c. Chart – Scatter plot

- In the board toolbar, **Visualize > Chart** -> Configure the chart as below and click on save
  - Chart Type: Scatter plot
    - If you hover your cursor over each option, the tooltip will tell you the title of each one.
    - Note that the same Chart board can produce many different kinds of charts, such as bar, line, scatter, heat grid and pie charts.
  - X-Axis: avg_purity.
  - Y-Axis: Series 1: Select equipment_id and Unique count
  - Options: Set Sort order to X axis ascending

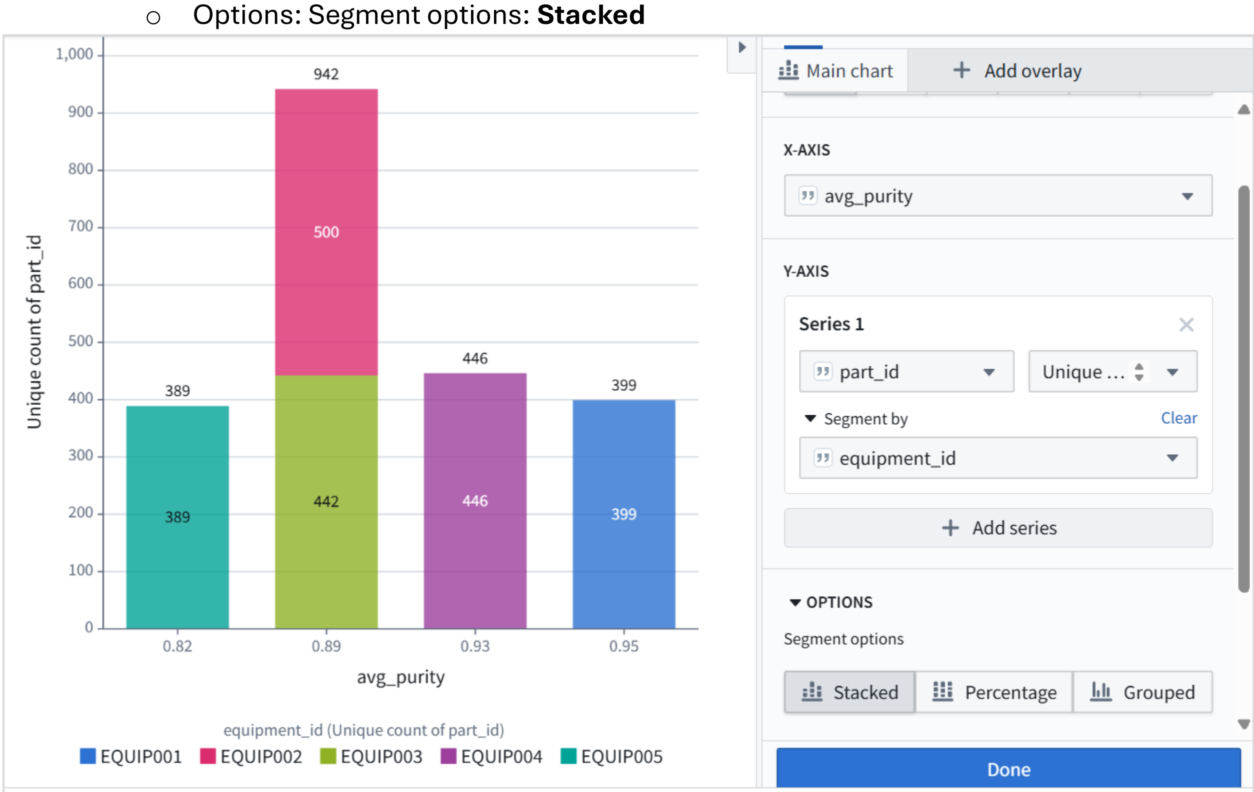
Again, you can hover over each option to reveal its title.



d. Chart – Vertical Bar Chart

➢ In the configuration of your existing chart / add a new chart board in the path and configure as follows and click on compute / save

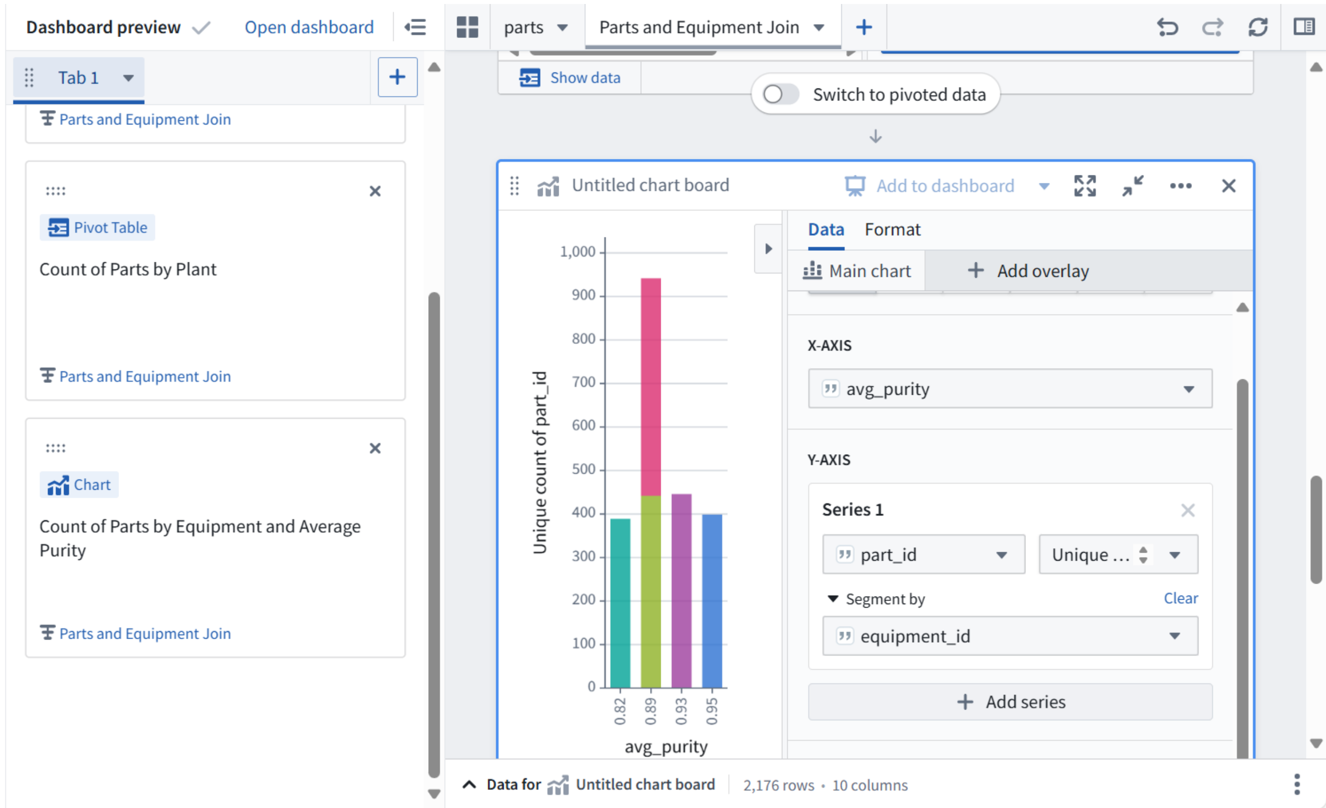
- Chart type: **Vertical bar chart**
- X-Axis: avg_purity
- Y-Axis: Series 1: Select part_id and **Unique count**
- Segment by: equipment_id
- Options: Segment options: **Stacked**



16. Contour Dashboard

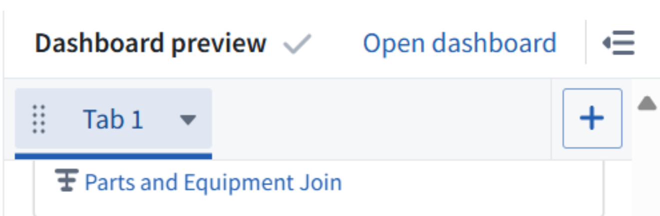
a. Create a dashboard of your analysis

1. Scroll to your Histogram board
2. Click **Add to dashboard** at the top of that board
   - The dashboard preview panel will appear on the left side of your screen and a blue *Dashboard* button will appear at the top.
3. Title this board: Inspection Alerts Histogram
   - The box in the center of the Histogram card in the dashboard preview allows you to title this board within the dashboard.
4. Add a title to your dashboard
   - Click into the text box at the top of the dashboard preview, which says *Untitled dashboard*
   - Enter: Inspection Prioritization Dashboard
5. Add your pivot table to the dashboard
   - Scroll to your pivot table board
   - Click **Add to dashboard**
   - Title the board: Count of Parts by Plant
6. Add your bar chart to the dashboard
   - Scroll to your pivot table board
   - Click **Add to dashboard**
   - Title the board: Count of Parts by Equipment and Average Purity



b. Interact with Dashboard

- Click **Open dashboard** at the top of the left dashboard panel

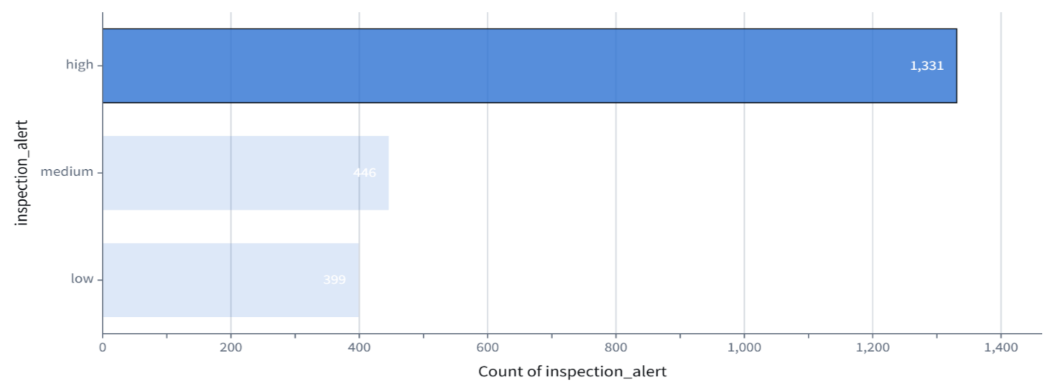


- In your dashboard, you may see the filters applied to each board as you selected or defined them in your analysis. These are interactive for viewers of the dashboard.
- Explore your dashboard parameters. Click the **(x)** to view the **Parameters**
- In the **Plant** filter, select **Plant 2**
- Click **Apply** at the bottom of the Parameters panel
  - The boards in the dashboard will dynamically update based upon your selection.
- Clear the parameter
- Click the **X** in the **Plant** text field to remove the list of plants -> click apply

Inspection Prioritization Dashboard

Tab 1 | + New tab

Inspection Alerts Histogram



Keep rows where inspection_alert is high — Overridden

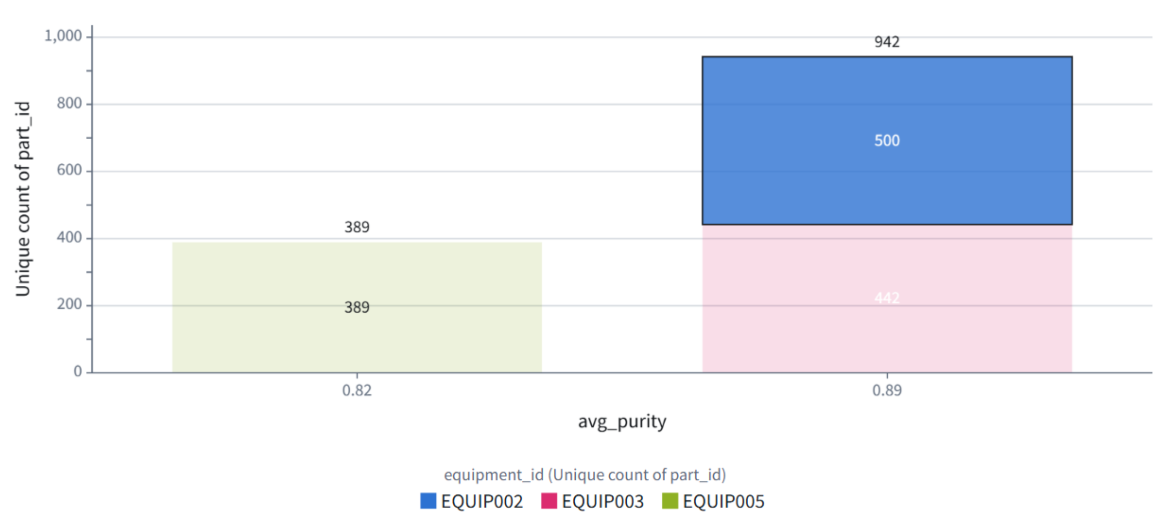
Show data — Affects 2 boards

Count of Parts by Plant

Displaying 3 rows

equipment_plant	equipment_id	count_distinct_of_part_
Plant 2	EQUIP002	500
Plant 3	EQUIP003	442
Plant 5	EQUIP005	389

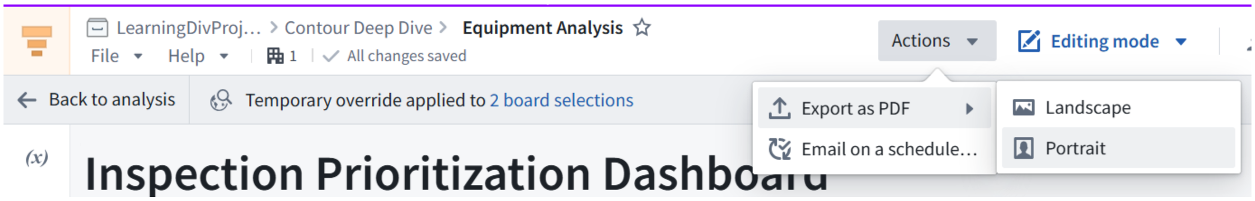
Count of Parts by Equipment and Average Purity



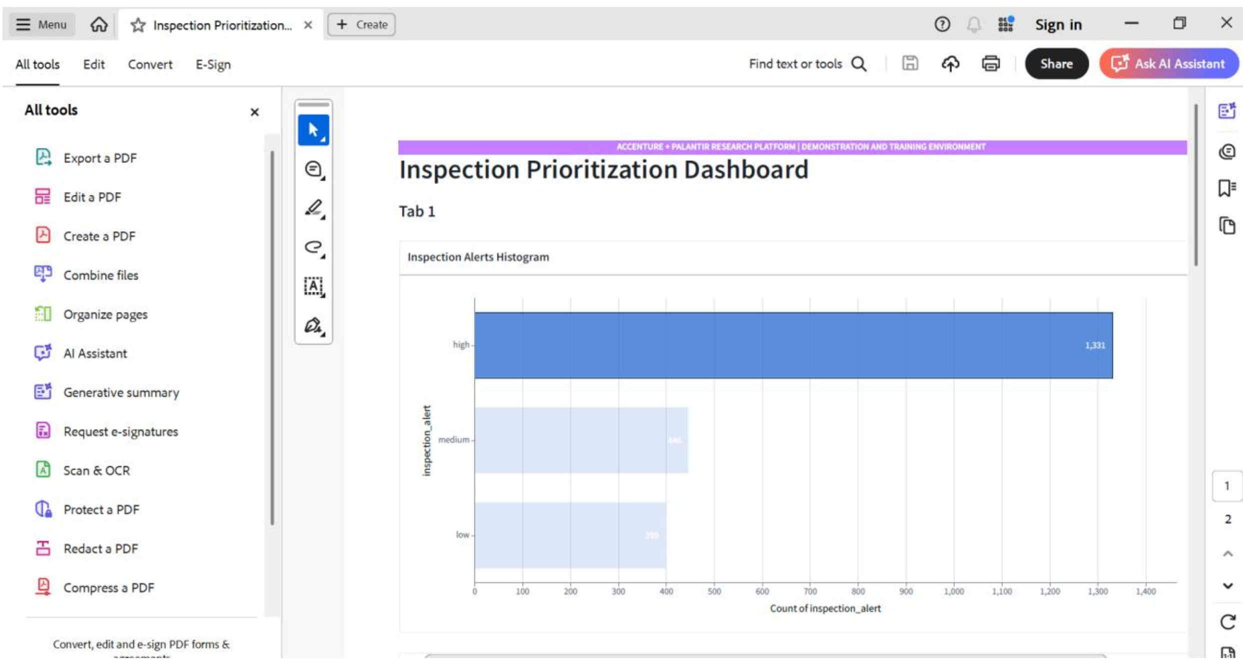
C. Export the Dashboard

- In the top right of the screen, click Actions > Export as PDF > Portrait
- It will take Foundry a few seconds to generate the PDF.

Any filters and overrides applied in the dashboard are included in the PDF report.



Open the exported PDF.



-- Happy Learning! --